

The Jet–Disk Connection in Thin Disks

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Abstract

We develop an analytic framework for jet suppression in radiatively efficient accretion disks around spinning black holes, extending the scale-invariant schematic model of Garofalo & Singh (2016). In that work, geometrically thin disks were argued to suppress relativistic jets by inhibiting the mass loading of the large-scale magnetic field structure associated with the Blandford–Payne effect. Here we turn that schematic picture into a spin-dependent analytic model by treating the disk atmosphere as a loading reservoir that is depleted by radiatively driven disk winds. The Blandford–Payne effect is taken to represent the slower, matter-dominated outer jet component required to support, collimate, and stabilize the inner relativistic jet powered by the Blandford–Znajek mechanism. Because centrifugal loading along inclined magnetic field lines requires plasma attachment to the disk atmosphere, the strength of the Blandford–Payne component depends sensitively on the density of material available above the disk surface. We show that as black hole spin becomes increasingly prograde, the inward shift of the ISCO increases the radiative efficiency of the thin disk and enhances radiative mass loss from the disk atmosphere. This depletes the plasma reservoir available for Blandford–Payne loading, weakening the outer magnetohydrodynamic jet structure even when comparable magnetic flux threads the black hole. The resulting suppression is therefore indirect. Radiative winds do not eliminate the Blandford–Znajek mechanism itself, but prevent the formation of the mass-loaded outer jet required for a powerful large-scale relativistic outflow. The model naturally favors strong jets in retrograde accretion systems and progressively weaker jets toward high prograde spin, offering a physical explanation for the radio loud/radio quiet dichotomy in active galactic nuclei and for jet suppression in radiatively efficient black hole accretion states.

1. Introduction

The production of relativistic jets in black hole accretion systems remains one of the central unsolved problems in high-energy astrophysics. Observations across the black hole mass scale reveal a strong connection between accretion state and jet production, with geometrically thick, radiatively inefficient accretion flows generally associated with powerful jets and geometrically thin, radiatively efficient disks associated with jet suppression (Fender et al. 2004; Marscher et al. 2002; Hardcastle et al. 2007; KÖrding et al. 2006). In active galactic nuclei (AGN), this distinction appears in the radio loud/radio quiet dichotomy, where only a minority of quasars produce powerful relativistic jets despite evidence that rapidly spinning black holes may exist in both radio loud and radio quiet systems (Sikora et al. 2007; Reynolds 2014). Similar behavior is observed in X-ray binaries, where transitions into radiatively efficient soft states are accompanied by strong jet suppression (Fender et al. 1999, 2004). These observations suggest that jet production depends not only on black hole spin, but also on how the accretion flow supplies plasma to the magnetic structures required for jet formation.

Theoretical models for relativistic jet production have historically emphasized large-scale magnetic fields threading rotating black holes and accretion disks. The Blandford–Znajek mechanism extracts rotational energy electromagnetically from the black hole through magnetic flux threading the event horizon, while the Blandford–Payne effect describes centrifugal acceleration of matter along inclined magnetic field lines anchored in the accretion disk (Blandford & Znajek 1977; Blandford & Payne 1982). General relativistic magnetohydrodynamic simulations have demonstrated the ability of magnetized accretion flows to produce relativistic jets, particularly in magnetically arrested disk states characterized by strong vertical magnetic flux accumulation near the black hole (Tchekhovskoy et al. 2011; McKinney et al. 2012; Narayan et al. 2022). However, despite major progress in numerical modeling, the physical origin of jet suppression in radiatively efficient thin disks remains poorly understood.

One common explanation for the radio loud/radio quiet dichotomy invokes differences in black hole spin, with radio loud AGN associated preferentially with rapidly spinning black holes (Sikora et al. 2007). However, X-ray reflection spectroscopy indicates that many radio quiet Seyferts and quasars may also host rapidly spinning black holes (Reynolds 2014; Walton et al. 2013; Brenneman 2013). If rapidly spinning black holes exist in both radio loud and radio quiet systems, then spin alone cannot determine whether a powerful large-scale jet forms. A second class of models emphasizes differences in magnetic flux accumulation, especially in magnetically arrested disk scenarios where powerful jets require large concentrations of vertical magnetic flux near the event horizon (Narayan et al. 2003; Tchekhovskoy et al. 2011). Yet differences in magnetic flux in otherwise similar accretion systems remain unclear.

In the present work, we adopt a different starting point. We assume that the large-scale magnetic flux threading the black hole and inner disk can be broadly comparable across the spin range. Therefore, the relevant spin-dependent control parameter is not primarily the amount of magnetic flux, nor the mere ability of field lines to bend into a Blandford–Payne favorable configuration. Field-line inclination remains a necessary condition for centrifugal launching, but it is not by itself sufficient. The field lines must also acquire plasma from the disk atmosphere. The central issue addressed here is therefore the availability of matter for mass loading, not the existence of magnetic flux or field-line bending alone.

Garofalo & Singh (2016) proposed a scale-invariant framework in which jet suppression emerges naturally in radiatively efficient accretion states through the interaction between thin-disk accretion and jet-supporting outflows. In that schematic picture, thin disks suppress jets not because of insufficient black hole spin or reduced magnetic flux, but because efficient radiative disks inhibit the mass loading required for the large-scale magnetohydrodynamic outflow associated with the Blandford–Payne effect. The present work builds directly on that framework by developing the underlying physics analytically and by explicitly modeling the competition between radiative atmospheric depletion and centrifugal mass loading as a function of black hole spin.

A key ingredient in the model is the dependence of radiative efficiency on black hole spin. As the ISCO radius decreases toward high prograde spin, the inner disk becomes increasingly radiatively efficient, enhancing radiatively driven mass loss from the disk atmosphere. This depletes the plasma reservoir available for Blandford–Payne loading. By contrast, high-spin retrograde

accretion is associated with a larger ISCO and weaker radiative disk winds, allowing the disk atmosphere to remain sufficiently dense for efficient mass loading. Thus, even if the magnetic flux and basic field-line geometry are comparable across spin, the ability of the disk to supply plasma to the outer MHD jet is strongly spin dependent.

The suppression mechanism therefore operates indirectly. Radiative winds do not need to eliminate the Blandford–Znajek mechanism or remove the magnetic flux threading the black hole. Instead, they suppress the mass-loaded Blandford–Payne component that forms the outer jet sheath required to support, collimate, and stabilize the inner relativistic jet. In this framework, differences in radio loudness emerge primarily from the efficiency of mass loading in the disk atmosphere, not from spin or magnetic flux alone.

This paper is organized as follows. In Section 2, we review the basic physics of centrifugal mass loading and clarify why field-line inclination is necessary but not sufficient for the Blandford–Payne component to operate efficiently. In Section 3, we derive a spin-dependent analytic model for radiative atmospheric depletion in thin accretion disks. Section 4 develops the competition between radiative mass loss and Blandford–Payne loading as a mechanism for jet suppression. In Section 5, we discuss implications for the radio loud/radio quiet dichotomy, state transitions in X-ray binaries, the limitations of current GRMHD simulations in modeling thin-disk jet suppression physics and conclude.

2. The Jet–Disk Connection and Blandford–Payne Mass Loading

The production of large-scale relativistic jets in black hole accretion systems is generally understood to require the combined action of rotating black holes, large-scale magnetic fields, and plasma capable of loading those magnetic structures. In the standard picture, the Blandford–Znajek mechanism (Blandford & Znajek 1977) extracts rotational energy electromagnetically from a spinning black hole through magnetic flux threading the event horizon, while the Blandford–Payne effect (Blandford & Payne 1982) centrifugally accelerates matter outward along inclined magnetic field lines anchored in the accretion disk. In the schematic developed in Garofalo & Singh (2016), the Blandford–Payne component is identified with the slower, matter-dominated outer jet structure that supports, collimates, and stabilizes the inner relativistic Blandford–Znajek jet. In this interpretation, efficient relativistic jet production requires not only electromagnetic extraction from the black hole, but also successful mass loading of the outer magnetohydrodynamic flow.

The essential condition for centrifugal launching in the Blandford–Payne effect is that the magnetic field lines emerging from the disk surface possess sufficient inclination relative to the vertical direction. For cold magnetocentrifugal acceleration, the poloidal magnetic field line must bend away from the vertical by more than 30° , or equivalently make an angle smaller than 60° relative to the disk plane (Blandford & Payne 1982). In cylindrical coordinates, this condition may be written approximately as

$$\frac{B_r}{B_z} > \frac{1}{\sqrt{3}}$$

where B_r and B_z are the radial and vertical magnetic field components evaluated near the disk surface. Once this condition is satisfied, matter attached to the magnetic field experiences an

outward centrifugal force along the field line, allowing plasma to escape the gravitational potential of the black hole–disk system.

A key point often overlooked is that magnetic inclination alone is insufficient for efficient outflow production. The field lines must also acquire plasma from the disk atmosphere. In other words, successful jet formation requires both magnetic geometry and sufficient surface density available for mass loading. The mass flux entering the outer MHD jet may be approximated schematically as

$$\dot{M}_{BP} \sim 2\pi r \rho_s c_s$$

where r is the characteristic launching radius, ρ_s is the density in the surface layers of the disk atmosphere, and c_s is the local sound speed. Although simplified, this expression captures the central physical point relevant for the present work. The efficiency of the outer jet depends directly on the availability of plasma near the disk surface.

In many jet models, differences between radio loud and radio quiet systems are attributed primarily either to black hole spin or to variations in magnetic flux accumulation near the event horizon. In the present framework, however, we assume that large-scale magnetic flux threading the black hole remains broadly comparable across the spin range. The dominant difference instead emerges from the ability of the accretion disk atmosphere to supply plasma to the outer magnetohydrodynamic jet structure. This shifts the focus of the jet problem away from magnetic flux accumulation alone and toward the interaction between the disk atmosphere and radiative disk physics.

Radiatively efficient thin disks produce substantial radiation pressure in their inner regions, particularly at high prograde black hole spin where the ISCO radius becomes small and the radiative efficiency increases. The enhanced radiation field naturally drives strong disk winds and atmospheric mass loss. In the context of the present model, these radiative winds compete directly with the centrifugal mass-loading process required for the Blandford–Payne component. Plasma removed from the disk atmosphere by radiative driving is no longer available to load the inclined magnetic field lines responsible for supporting the outer jet.

The competition between these two processes forms the basis of the jet suppression mechanism explored here. The essential idea is not that radiative winds dynamically destroy an already-formed relativistic jet. Rather, radiative atmospheric depletion suppresses the formation of the mass-loaded outer MHD jet structure required to sustain the large-scale Blandford–Znajek outflow. As radiative efficiency increases toward high prograde spin, progressively less plasma remains available for centrifugal loading, weakening the outer jet and indirectly suppressing the large-scale relativistic jet despite continued electromagnetic extraction near the black hole horizon.

This picture naturally explains why rapidly spinning black holes need not always produce powerful relativistic jets. Even if magnetic flux threading the black hole remains substantial, insufficient plasma loading of the outer magnetohydrodynamic component may prevent the formation of a stable, well-collimated large-scale outflow. In the following section, we develop a

spin-dependent analytic model for the competition between radiative atmospheric depletion and Blandford–Payne mass loading in thin accretion disks.

3. Spin-Dependent Radiative Atmospheric Depletion

The efficiency of radiative mass loss from an accretion disk depends strongly on the structure of the inner flow and therefore on black hole spin. In geometrically thin, optically thick accretion disks, the location of the innermost stable circular orbit (ISCO) determines both the gravitational binding energy available for radiation and the radial concentration of energy dissipation within the disk. Because the ISCO moves inward with increasing prograde spin, radiatively efficient accretion flows around rapidly spinning prograde black holes produce substantially larger radiative fluxes in their inner regions than retrograde systems.

The general relativistic Novikov–Thorne thin-disk flux (or dissipation) function is written as

$$F(r) = \frac{3GM\dot{M}_{acc}}{8\pi r^3} f(r,a),$$

where $\dot{M}_{acc}$ is the accretion rate and $f(r,a)$ is the relativistic correction factor which is expressed as:

$$f(r,a) = \frac{x^2}{x^3 - 3x + 2a} \left[x - x_0 - \frac{3}{2} a \ln\left(\frac{x}{x_0}\right) - \sum_{i=1}^3 \frac{3(x_i - a)^2}{x_i(x_i - x_j)(x_i - x_k)} \ln\left(\frac{x - x_i}{x_0 - x_i}\right) \right]$$

where

$$x = \sqrt{\frac{r}{r_g}}, x_0 = \sqrt{\frac{r_{ISCO}}{r_g}},$$

r_g is the gravitational radius and x_i are the three roots of

$$x^3 - 3x + 2a = 0.$$

These roots are analytically evaluated as:

$$x_1 = 2\cos\left(\frac{1}{3}\cos^{-1}a - \frac{\pi}{3}\right), x_2 = 2\cos\left(\frac{1}{3}\cos^{-1}a + \frac{\pi}{3}\right), x_3 = -2\cos\left(\frac{1}{3}\cos^{-1}a\right).$$

We evaluate the peak value of $F(r)$ and approximate it analytically as follows, which is normalized to its highest value at spin of 0.998.

$$F(a) = \text{constant} \times 10^{-2.75+2.8a+0.15a^2}.$$

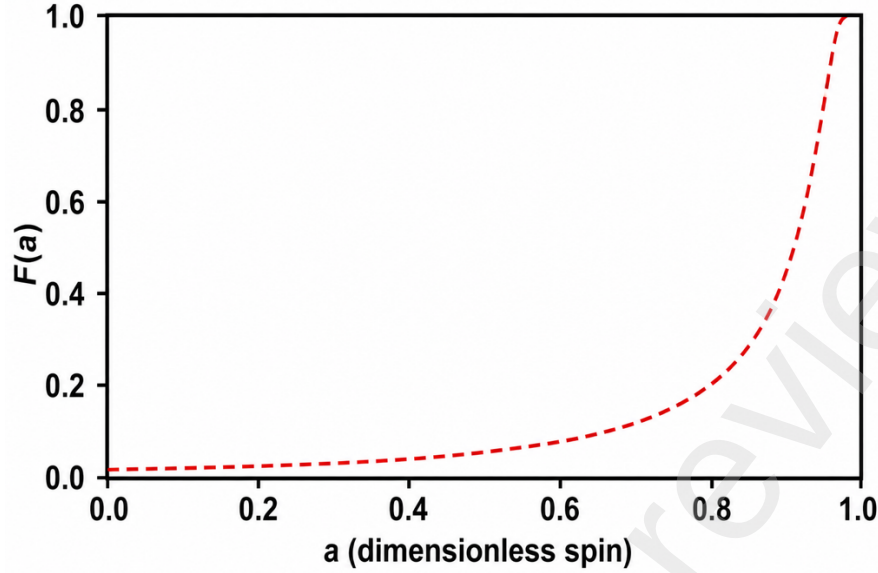


Figure 1: Dissipation function peak value as a function of black hole spin.

The function $F(a)$ increases monotonically with prograde black hole spin and becomes strongly nonlinear above $a \sim 0.7 - 0.8$, reflecting the onset of the BP suppression regime. The second derivative of the dissipation function, d^2F/da^2 , provides a measure of how rapidly the sensitivity of the inner-disk radiation field changes with black hole spin. For the analytic approximation the second derivative is

$$\frac{d^2F}{da^2} = \text{constant} \times F(a) [0.3 \ln(10) + (2.8 + 0.3a)^2 (\ln 10)^2].$$

While the first derivative $\frac{dF}{da}$ quantifies the rate at which the radiative dissipation increases with prograde spin, the second derivative measures the acceleration of that increase. Numerically, $\frac{d^2F}{da^2}$ rises from approximately 0.07 at $a = 0$ to ~ 12 at $a = 0.8$, reaching values of order ~ 45 near maximal prograde spin ($a = 0.998$). The dissipation function therefore enters a strongly nonlinear regime near $a \sim 0.7 - 0.9$. Physically, this behavior reflects the rapid inward collapse of the ISCO in Kerr spacetime, which concentrates gravitational energy release into progressively smaller radii and dramatically enhances the local radiation field. In the present framework, this nonlinear increase is important because radiatively driven atmospheric depletion depends directly on the strength of the inner-disk dissipation. The sharp rise in $\frac{d^2F}{da^2}$ therefore naturally identifies a transition region above which radiative disk winds become increasingly effective at removing plasma from the disk atmosphere, thereby suppressing Blandford–Payne mass loading and indirectly weakening large-scale relativistic jet production.

The corresponding radiative luminosity of the disk is given by

$$L = \eta(a) M_{acc}^2 c$$

where $\eta(a)$ is the spin-dependent accretion efficiency. Typical efficiencies range from $\eta \sim 0.04$ for maximally retrograde accretion to $\eta \sim 0.3$ for near-maximal prograde spin. The resulting increase in radiative power has important consequences for the structure of the disk atmosphere.

Radiative acceleration acting on plasma near the disk surface may be approximated as

$$a_{rad} \sim \frac{\kappa F}{c}$$

where κ is the opacity and F is the local radiative flux. As the radiative flux increases toward high prograde spin, radiation pressure becomes increasingly effective at removing plasma from the surface layers of the disk. This process generates radiatively driven winds and progressively depletes the atmospheric material available for centrifugal loading onto inclined magnetic field lines.

The characteristic radiative mass-loss rate may be estimated from momentum conservation. Equating the radiative momentum flux with the momentum carried by the outflow yields

$$\dot{M}_{rad} v_{esc} \sim \frac{L}{c}$$

or equivalently

$$\dot{M}_{rad} \sim \frac{L}{c v_{esc}}$$

where v_{esc} is the characteristic escape velocity near the launching region. Substituting the thin-disk luminosity relation gives

$$\dot{M}_{rad}(a) \propto \eta(a)$$

to first approximation. Thus, the efficiency of radiative atmospheric stripping increases systematically toward high prograde spin.

The key physical consequence is that the density in the disk atmosphere becomes spin dependent. If ρ_s denotes the characteristic surface density available for centrifugal loading, we may schematically write

$$\rho_s(a) \sim \rho_0 - k \dot{M}_{rad}(a)$$

where ρ_0 is the atmospheric density in the absence of radiative depletion and k parameterizes the efficiency with which radiative winds remove material from the disk surface layers. Since $\dot{M}_{rad}$ increases strongly toward high prograde spin, the surface density available for mass loading correspondingly decreases.

This effect is especially important in the inner regions of the disk where the Blandford–Payne component is expected to couple most efficiently to the relativistic jet. In retrograde systems, the larger ISCO radius reduces both the radiative efficiency and the concentration of inner-disk radiation, allowing substantial plasma to remain near the disk surface. Efficient centrifugal

loading can therefore occur along inclined magnetic field lines, sustaining the slower matter-dominated outer jet required to support the relativistic Blandford–Znajek outflow. In contrast, high prograde systems generate strong radiative atmospheric depletion that progressively suppresses plasma attachment to the field lines.

The important point is that the suppression mechanism does not require any reduction in magnetic flux threading the black hole. In the present framework, magnetic flux may remain substantial across the full spin range. Instead, the key quantity controlling large-scale jet production is the availability of plasma capable of loading the outer magnetohydrodynamic structure. As the atmospheric density decreases toward high prograde spin, the efficiency of centrifugal loading declines correspondingly, weakening the outer jet and indirectly suppressing the large-scale relativistic outflow.

In the next section, we combine the spin-dependent radiative depletion model developed here with the centrifugal loading conditions of Section 2 in order to derive an analytic criterion for jet suppression in radiatively efficient accretion disks.

4. Jet Suppression Through Radiative Depletion of Blandford–Payne Loading

The central mechanism proposed here is that relativistic jet suppression in radiatively efficient accretion systems is controlled by the density of the disk atmosphere available for Blandford–Payne loading. In the framework of Garofalo & Singh (2016), the Blandford–Payne effect supplies the slower, matter-dominated outer jet component that supports, collimates, and stabilizes the inner relativistic jet powered by the Blandford–Znajek mechanism. The present work makes this picture quantitative by treating the disk atmosphere as a mass reservoir that can either supply plasma to inclined magnetic field lines or be depleted by radiatively driven disk winds.

Let $\rho_s(a)$ denote the density of the disk atmosphere available for centrifugal loading at a characteristic radius r . In the absence of radiative depletion, this density is ρ_0 . Radiative winds remove material from this same atmospheric reservoir at a spin-dependent rate $M_{rad}(a)$. We therefore write the available surface density schematically as

$$\rho_s(a) = \rho_0 \max \left[0, 1 - \frac{M_{rad}(a)}{M_{dep}} \right],$$

where M_{dep} is the characteristic radiative mass-loss rate required to substantially deplete the disk surface layers. This form captures the essential physical behavior. When radiative depletion is weak, $\rho_s(a) \approx \rho_0$; however, as radiative depletion approaches the depletion threshold, the atmospheric density available for Blandford–Payne loading collapses. We obtain

$$M_{rad}(a) \sim \frac{\eta(a) M_{acc} c}{v_{esc}}.$$

For a characteristic loading radius r ,

$$v_{esc}(r) \sim \left(\frac{2GM}{r}\right)^{1/2},$$

so that

$$M_{rad}(a) \sim \eta(a)M_{acc}\left(\frac{r}{2r_g}\right)^{1/2},$$

where $r_g = GM/c^2$. The spin dependence enters primarily through the radiative efficiency $\eta(a)$, which increases strongly toward high prograde spin as the ISCO moves inward. Substituting this into the atmospheric reservoir model gives

$$\rho_s(a) = \rho_0 \max\left[0, 1 - \frac{\eta(a)M_{acc}\left(\frac{r}{2r_g}\right)^{1/2}}{M_{dep}}\right].$$

This expression is the core of the suppression model. It states that the plasma density available for Blandford–Payne loading decreases as the radiative efficiency increases. Retrograde accretion, with a large ISCO and relatively low radiative efficiency, produces weak atmospheric depletion and preserves a dense loading reservoir. High-spin prograde accretion, with a small ISCO and high radiative efficiency, produces strong radiative mass loss and depletes the atmosphere from which the Blandford–Payne component must be loaded.

The corresponding mass flux into the Blandford–Payne outer jet yields

$$M_{BP}(a) \sim 2\pi r c_s \rho_0 \max\left[0, 1 - \frac{\eta(a)M_{acc}\left(\frac{r}{2r_g}\right)^{1/2}}{M_{dep}}\right].$$

Thus, even if the magnetic flux threading the black hole and inner disk remains comparable across the spin range, the mass-loaded outer jet is strongly spin dependent because the atmospheric reservoir is spin dependent. The suppression is therefore not primarily a failure of magnetic flux accumulation. It is a failure of plasma supply to the inclined magnetic field lines that produce the Blandford–Payne outer jet.

A useful dimensionless measure of the competition is

$$\Lambda(a) \equiv \frac{M_{BP}(a)}{M_{rad}(a)}.$$

Efficient large-scale jet production occurs when

$$\Lambda(a) \gg 1,$$

because the disk atmosphere supplies plasma to the Blandford–Payne component faster than radiation removes it. Jet suppression occurs when

$$\Lambda(a) \lesssim 1,$$

or equivalently when radiative depletion dominates over centrifugal loading. In terms of the atmospheric reservoir model, the stronger condition for severe suppression is

$$\frac{\eta(a)M_{acc}\left(\frac{r}{2r_g}\right)^{1/2}}{M_{dep}} \gtrsim 1.$$

At this point the disk atmosphere is substantially depleted and $\rho_s(a)$ approaches zero. The Blandford–Payne mass-loading rate then collapses, leaving the inner Blandford–Znajek jet without the slower matter-dominated outer sheath required for stable collimation and large-scale propagation.

This framework gives a natural spin-dependent interpretation of jet suppression. In high-spin retrograde systems, the larger ISCO reduces the radiative efficiency and weakens disk winds. The atmosphere remains dense, $M_{rad} \ll M_{BP}$, and the Blandford–Payne component thrives. The result is a strong outer MHD jet capable of supporting a powerful large-scale relativistic outflow. In high-spin prograde systems, the smaller ISCO increases the radiative efficiency and strengthens radiative disk winds. The atmosphere is depleted, $M_{rad} \gtrsim M_{BP}$, and the Blandford–Payne component fails. The inner Blandford–Znajek mechanism may still operate locally, but without the outer mass-loaded sheath the large-scale jet is weak, poorly collimated, or absent.

The important point is that jet suppression is indirect. Radiative winds do not need to destroy a fully formed relativistic jet. Instead, they prevent the formation of the outer Blandford–Payne jet by removing the plasma required for centrifugal loading. In this sense, the disk wind suppresses the jet by starving the jet-supporting component at its origin. This provides a physical route to radio quiet, high-spin systems without requiring either low black hole spin or negligible magnetic flux.

5. Discussion and Conclusions

The framework developed in this paper has important implications for the interpretation of jet production across black hole accretion systems. Because the suppression mechanism is tied fundamentally to the interaction between radiative disk physics and centrifugal mass loading, the model naturally connects the radio loud/radio quiet dichotomy in AGN with accretion state transitions in X-ray binaries within a unified scale-invariant picture. In both cases, the essential physics depends not primarily on black hole mass, but on the ability of the accretion flow to maintain a sufficiently dense disk atmosphere capable of loading the outer magnetohydrodynamic jet component.

In AGN, the model predicts that the most powerful large-scale relativistic jets should preferentially occur in systems with relatively weak radiative atmospheric depletion. This condition is naturally satisfied in retrograde accretion systems because the larger ISCO radius reduces both radiative efficiency and the concentration of radiation emerging from the inner disk. Substantial plasma therefore remains available for centrifugal loading onto inclined magnetic field lines, sustaining the outer MHD structure required for stable Blandford–Znajek jet propagation. As the accretion geometry evolves toward increasingly prograde configurations,

radiative efficiency increases and atmospheric depletion becomes progressively more effective, leading naturally to weaker jets and eventually to radio quiet states despite substantial black hole spin.

This interpretation is consistent with the growing observational evidence that rapidly spinning black holes exist in both radio loud and radio quiet systems (Reynolds 2014; Brenneman 2013). In conventional spin paradigms, the existence of high-spin radio quiet AGN is difficult to reconcile with the expectation that large spin should produce strong relativistic jets through efficient electromagnetic extraction. In the present framework, however, high spin simultaneously increases radiative atmospheric stripping, thereby reducing the efficiency of centrifugal loading in the outer jet. Large black hole spin therefore becomes compatible with jet suppression once the role of disk atmosphere depletion is included.

The model also provides a natural explanation for the observed anti-correlation between strong disk winds and large-scale relativistic jets in some AGN populations. Radiatively efficient systems exhibiting strong winds are precisely those expected to experience substantial atmospheric depletion. In such systems, plasma removal from the disk surface suppresses the formation of the outer magnetohydrodynamic jet component required to support the relativistic outflow. Conversely, systems with weaker radiative winds retain sufficient atmospheric plasma for efficient centrifugal loading and stronger jet production. The jet–wind connection therefore emerges naturally from the competition between radiative depletion and Blandford–Payne loading.

The same physical framework extends naturally to X-ray binaries. During hard states, accretion flows are geometrically thick and radiatively inefficient, producing relatively weak radiative atmospheric depletion and strong steady jets. During transitions into soft states, however, the accretion disk becomes geometrically thin and radiatively efficient. The resulting increase in radiative atmospheric stripping suppresses centrifugal loading of the outer magnetohydrodynamic jet component, leading to the well-known disappearance of steady relativistic jets in soft states (Fender et al. 1999, 2004). The observed scale invariance between AGN and X-ray binaries therefore follows directly from the underlying disk atmosphere physics rather than from black hole mass itself.

An important implication of the present model concerns the interpretation of GRMHD simulations. Modern simulations have successfully demonstrated the ability of magnetized accretion flows to generate relativistic jets, particularly in magnetically arrested disk configurations characterized by strong vertical magnetic flux accumulation near the event horizon (Tchekhovskoy et al. 2011; McKinney et al. 2012). However, many numerical studies focus primarily on the production of relativistic jets rather than on the physical origin of jet suppression in radiatively efficient thin disks. The present framework suggests that this limitation may arise because the key suppression physics depends sensitively on the thermodynamic structure of the disk atmosphere and on the competition between radiative winds and centrifugal mass loading.

In particular, radiative atmospheric depletion requires self-consistent treatment of thin-disk radiation physics, radiative transfer, opacity-dependent acceleration, and disk surface thermodynamics. These processes remain difficult to model in current GRMHD calculations,

especially in geometrically thin disks where numerical resolution requirements become severe. Furthermore, many simulations begin with large-scale magnetic field configurations designed specifically to maximize jet production, potentially obscuring the conditions under which centrifugal mass loading fails. The suppression mechanism proposed here therefore highlights the need for future numerical work capable of simultaneously modeling radiatively efficient thin disks, radiative atmospheric stripping, and magnetically loaded outer jet formation.

More generally, the framework developed in this paper suggests that the jet problem may fundamentally be a disk atmosphere problem. In this picture, the critical quantity governing large-scale jet formation is not simply black hole spin or magnetic flux accumulation, but rather the ability of the disk atmosphere to maintain sufficient plasma for centrifugal loading of the outer magnetohydrodynamic structure. The jet–disk connection therefore emerges from the competition between electromagnetic extraction, radiative disk physics, and atmospheric mass supply.

DATA AVAILABILITY STATEMENT

The data is in the paper

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